

Practical 8

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

```
#include <stdio.h>

int main() {
    int frames[10], time[10];
    int n, f, i, j, page, flag, pos;
    int faults = 0, hits = 0, counter = 0;

    printf("Enter number of frames: ");
    scanf("%d", &f);

    printf("Enter number of pages: ");
    scanf("%d", &n);

    int pages[n];
    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &pages[i]);

    for(i = 0; i < f; i++) {
        frames[i] = -1;
        time[i] = 0;
    }

    for(i = 0; i < n; i++) {
        page = pages[i];
        flag = 0;

        for(j = 0; j < f; j++) {
            if(frames[j] == page) {
                hits++;
                counter++;
                time[j] = counter;
                flag = 1;
                break;
            }
        }

        if(flag == 0) {
            faults++;
            pos = 0;
            for(j = 1; j < f; j++) {
                if(time[j] < time[pos])
                    pos = j;
            }
            frames[pos] = page;
            counter++;
            time[pos] = counter;
        }
    }
}
```

```

        if(time[j] < time[pos])
            pos = j;
    }
    frames[pos] = page;
    counter++;
    time[pos] = counter;
}

printf("\nTotal Page Hits: %d", hits);
printf("\nTotal Page Faults: %d", faults);
printf("\nHit Percentage: %.2f%%", (hits * 100.0) / n);
printf("\nFault Percentage: %.2f%%\n", (faults * 100.0) / n);

return 0;
}

```

```

sh@DESKTOP-0TK9CH3:~$ nano lru.c
sh@DESKTOP-0TK9CH3:~$ gcc lru.c -o lru
sh@DESKTOP-0TK9CH3:~$ ./lru
Enter number of frames: 3
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5

Total Page Hits: 2
Total Page Faults: 10
Hit Percentage: 16.67%
Fault Percentage: 83.33%
sh@DESKTOP-0TK9CH3:~$ |

```